

AI MSP Service Desk

Automation Platform

Complete Technical Documentation

Built with n8n • GPT-4o-mini • Google Sheets • Slack • Gmail

1. Project Overview

The AI MSP Service Desk is an end-to-end support ticket automation platform built in n8n. It receives customer support requests, uses AI to classify and triage them, checks them against a curated knowledge base, and either resolves them automatically or escalates them to a human engineer. High-priority tickets require explicit human approval in Slack before any customer communication is sent.

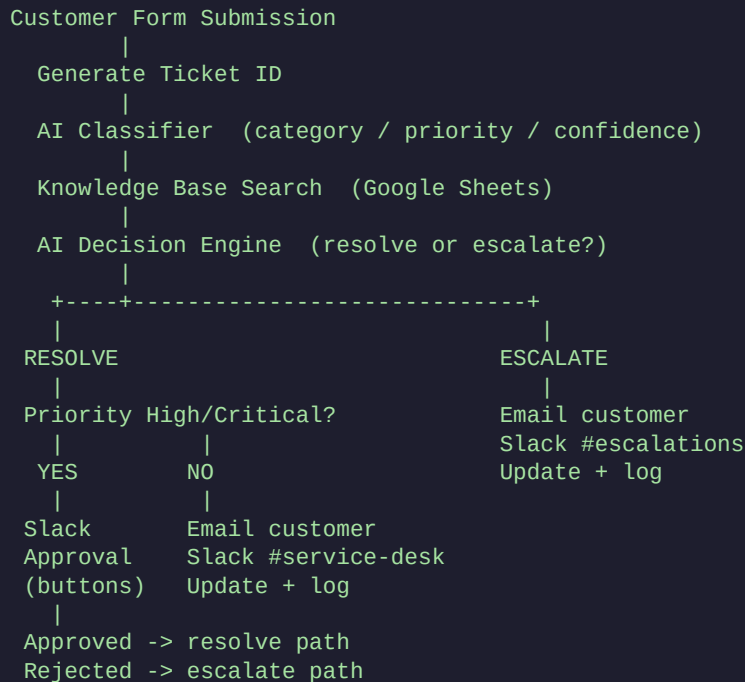
The system is designed to eliminate repetitive Level 1 support work while keeping a human in control of anything high-stakes. Each auto-resolved ticket saves a technician an estimated 10 to 15 minutes of manual effort, which adds up to over 12 hours of support time saved in a typical week.

Key Capabilities

- AI-powered ticket classification with category, priority, and confidence scoring
- Knowledge-base-grounded resolution so no hallucinated fixes reach customers
- Automated customer email for resolved tickets
- Human-in-the-loop approval via interactive Slack buttons for high-priority tickets
- Automatic escalation to the engineering team for anything the AI cannot safely resolve
- Full ticket and execution logging to Google Sheets (acts as a lightweight ticket database)
- Retry logic and a dedicated error-handling lane for API failures

2. Architecture at a Glance

A ticket flows through the system in a single n8n execution. The high-level path is:



Design principle

The AI can only resolve a ticket using a documented knowledge base solution.

If there is no clear KB match, the ticket is escalated to a human. This prevents the model from inventing technical instructions and sending them to customers.

3. Technology Stack

Component	Technology	Role
Workflow engine	n8n (self-hosted on Railway)	Orchestrates the entire ticket lifecycle
AI model	OpenAI GPT-4o-mini	Classification and resolve/escalate decisions
Knowledge base	Google Sheets	Stores documented Issue/Solution pairs
Ticket database	Google Sheets	Tickets and Logs tabs (HaloPSA-style DB)
Customer email	Gmail	Sends resolution and acknowledgment emails
Team notifications	Slack	Escalations, approvals, and error alerts

4. Credentials Reference

The workflow references these credentials by exact name. The names must match precisely in n8n, or the affected nodes will fail with a credential-not-found error.

Credential name	Type	Used by
OpenAi account	OpenAI API	AI Classifier and AI Decision Engine models
Google Sheets account 2	Google Sheets OAuth2	All 6 Google Sheets nodes (KB, Tickets, Logs)
Gmail account 4	Gmail OAuth2	Both customer email nodes
Slack account 2	Slack API	All 4 Slack nodes

5. Google Sheets Setup

Two separate spreadsheets are required.

Spreadsheet A — MSP Knowledge Base

One tab named exactly KnowledgeBase with two columns:

Column A	Column B
Issue	Solution
Outlook Authentication	Sign out of Outlook, clear cached credentials, sign back in...
VPN Connection Failed	Restart the VPN client and reinstall the profile if the certificate expired...

Spreadsheet B — MSP Ticket Database

Tab 1: Tickets

Columns A through K: Ticket ID, Status, Engineer, Priority, Company, Issue, Resolution, Assigned To, Created, Resolved, AI Summary.

Tab 2: Logs

Columns A through G: Timestamp, Workflow, Ticket ID, User, API Status, Response, Error.

Tab names are case-sensitive

The workflow looks for tabs named exactly KnowledgeBase, Tickets, and Logs.
A typo such as 'tickets' or 'Knowledge Base' will cause the Sheets nodes to fail.

6. Complete Node Reference

The workflow contains 27 functional nodes (plus documentation sticky notes). They are grouped by stage below.

Stage 1 — Intake and Classification

Node	Type	Function
Ticket Intake Form	Form Trigger	Hosts the customer form (Name, Company, Email, Problem). Entry point of the workflow.
Generate Ticket ID	Code	Creates a TKT-YYYY-NNNNNN ID, validates required fields, normalises the field names.
OpenAI Model - Classifier	OpenAI Chat Model	GPT-4o-mini language model feeding the classifier agent.
Classification Schema	Structured Output Parser	Forces the classifier output into a strict JSON schema.
AI Classifier	AI Agent	Classifies the ticket: category, priority, department, issue, confidence.
Normalize Classification	Code	Merges ticket data and AI output into one clean object.

Stage 2 — Knowledge Base and Decision

Node	Type	Function
Knowledge Base Search	Google Sheets	Reads all rows from the KnowledgeBase tab.
Merge KB Context	Code	Formats KB rows into readable text using a whitespace/case-tolerant column lookup.
OpenAI Model - Decision	OpenAI Chat Model	GPT-4o-mini language model feeding the decision agent.
Decision Schema	Structured Output Parser	Forces the decision output into a strict JSON schema.
AI Decision Engine	AI Agent	Decides resolve vs escalate based on KB match, confidence, category, and priority.
Can Be Resolved?	IF	Routes TRUE (resolve) or FALSE (escalate) on the AI decision action.

Stage 3 — Human Approval

Node	Type	Function
Priority Gate	IF	On the resolve path: checks if priority is High or Critical. If so, requires approval.
Slack - Approval Request	Slack (Send and Wait)	Posts an interactive Approve/Disapprove message and PAUSES the workflow until a manager responds.
Check Approval Decision	IF	Reads the approved boolean. TRUE continues to resolution; FALSE escalates.

Stage 4 — Resolution Path

Node	Type	Function
Email Customer - Resolved	Gmail	Emails the customer the resolution steps. 3 retries, error output active.
Slack - Notify Resolved	Slack	Posts a resolved notification to #service-desk.
Update Ticket DB - Resolved	Google Sheets	Upserts the Tickets row with Status: Resolved.
Log Execution - Resolved	Google Sheets	Appends a success row to the Logs tab.

Stage 5 — Escalation Path

Node	Type	Function
Email Customer - Escalated	Gmail	Tells the customer a technician will follow up. 3 retries, error output active.
Slack - Escalation Alert	Slack	Posts a red alert to #service-desk-escalations with full context.
Update Ticket DB - Escalated	Google Sheets	Upserts the Tickets row with Status: Escalated.
Log Execution - Escalated	Google Sheets	Appends an escalated row to the Logs tab.

Stage 6 — Completion and Error Handling

Node	Type	Function
Ticket Processing Complete	No Operation	Final convergence node for both resolve and escalate paths.
Error Handler - Email Failure	Code	Activates when all Gmail retries fail. Formats error metadata and backoff schedule.
Log Error to Sheets	Google Sheets	Appends an ERROR row to the Logs tab.
Slack - Error Alert	Slack	Alerts #service-desk-engineering for manual intervention.

7. AI Prompt Design

Two AI agents drive the workflow. Both are configured to return strict JSON and are given explicit, ordered rules to keep behaviour deterministic.

AI Classifier

Assigns category, priority, department, issue, and confidence. Priority is defined explicitly by scope and impact:

- Low: single user, no deadline, minor inconvenience
- Medium: single user blocked, or a routine request
- High: multiple users, a named deadline, or an executive blocked (not a full outage)
- Critical: company or department-wide outage, or an active security breach

AI Decision Engine

Applies these rules in order; the first match wins:

1. Escalate if classification confidence is below 70
2. Escalate if category is Security Alert, Hardware, or New Employee
3. Escalate if priority is Critical
4. Escalate if no knowledge base article clearly matches
5. Otherwise resolve, using resolution steps taken directly from the matching KB article

Grounding rule

When resolving, the resolution steps must come from the provided KB article.
The model is instructed never to add steps not supported by the knowledge base.

8. Production-Readiness Features

Retry Logic

Every external-service node (Gmail, Slack, Google Sheets) is configured with 3 retry attempts and a 30-second wait between tries. The two Gmail nodes additionally route to a dedicated error lane if all retries fail.

Error Handling Lane

When a customer email fails after all retries, the error output activates an Error Handler that logs the failure metadata (including a documented 30s/60s/120s backoff schedule) to the Logs sheet and alerts the engineering team in Slack, so no ticket is silently dropped.

Human-in-the-Loop Approval

High and Critical priority tickets pause at the Slack Approval Request node. Using n8n's native Send-and-Wait operation, a manager sees Approve and Disapprove buttons directly in Slack. The workflow does not send any customer communication until a human responds, and auto-escalates if no response arrives within the configured window.

Structured Logging

Every ticket writes to two places: the Tickets tab (current state of each ticket) and the Logs tab (an append-only audit trail of every execution event). Together these act as a lightweight ticket database.

9. Slack Configuration

The interactive approval buttons require Slack Interactivity to be enabled on the Slack app.

Required channels

Channel	Purpose
#service-desk	Auto-resolved ticket notifications
#service-desk-escalations	Tickets escalated to human engineers
#service-desk-approvals	High-priority approval requests (interactive buttons)
#service-desk-engineering	Email delivery failure alerts

Enabling Interactivity

6. Go to api.slack.com/apps and select your app
7. Under Features, open Interactivity & Shortcuts
8. Toggle Interactivity ON
9. Set the Request URL to: <https://YOUR-INSTANCE/webhook-waiting>
10. Save changes

Self-hosted requirement

For approval buttons and resume to work on a self-hosted instance, the `WEBHOOK_URL` environment variable must be set to your public instance URL, for example: `WEBHOOK_URL=https://your-instance.up.railway.app/`

10. Testing Guide

Test each route by submitting the form with these problem descriptions.

Auto-resolve (resolves immediately, no approval)

Test input	Matches
I can't connect to Outlook after changing my password	Outlook Authentication
My VPN won't connect, it just times out	VPN Connection Failed
The office printer isn't responding to print jobs	Printer Not Responding

Human approval (High priority + KB match)

Test input	Then
Five finance staff cannot access Microsoft 365 and cannot process invoices due today	Click Approve
Three sales members are locked out before a client demo this afternoon	Click Disapprove

Escalation

Test input	Reason
Someone is trying to log in from an unknown IP in another country	Security Alert category
My laptop screen is cracked and needs replacing	Hardware category
Our entire company has lost all internet	Critical priority

11. Troubleshooting

Symptom	Cause	Fix
Credential not found	Name mismatch	Rename credential to match exactly
Sheets node 'document not found'	Placeholder ID still in node	Set the real Sheet ID in the node
KB Solution shows 'undefined'	Header name mismatch	Ensure headers are Issue and Solution
Slack buttons do nothing	Interactivity not enabled	Enable Interactivity + set Request URL
Approval never pauses	WEBHOOK_URL not set	Set WEBHOOK_URL env var and redeploy
'Invalid token' on approval	Wrong resume base URL	Fix WEBHOOK_URL to public instance URL
Form submits, nothing happens	Workflow not active	Activate the workflow
Email goes to spam	No SPF/DKIM on domain	Configure SPF/DKIM for production

12. Business Impact

The value of the platform comes from removing manual effort on routine tickets while preserving human oversight where it matters.

Metric	Value
Time to resolve a routine ticket (manual)	10 to 15 minutes
Time to resolve via automation	Under 1 minute, zero human time
Estimated saving per auto-resolved ticket	10 to 15 minutes
Estimated weekly saving (typical volume)	Over 12 hours
Human oversight	Retained on all High/Critical tickets